

الجامعة المصرية اليابانية للعلوم و التكنولوجيا
エジプト日本科学技術大学
EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

ECE 212: ELECTRONICS ENGINEERING LAB

Experiment 6 - Bipolar Junction Transistor common emitter amplifier

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Section 3

Group 1

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Q1) Bias point detail simulation & experimental.

The table below shows the bias point detail in simulation & experiment.

As shown the values are almost the same.

value	Vcc	V_B	V_E	V_C	I_B	I_E	I_C	β	α
simulation	10	3.003V	2.346V	6.49V	3.43µA	1.066mA	1.063mA	309.91	0.9967
measured	10	3	2.278V	6.484V	3.6µA	1.036mA	1.06mA	377.77	0.9973

Q2) small signal analysis.

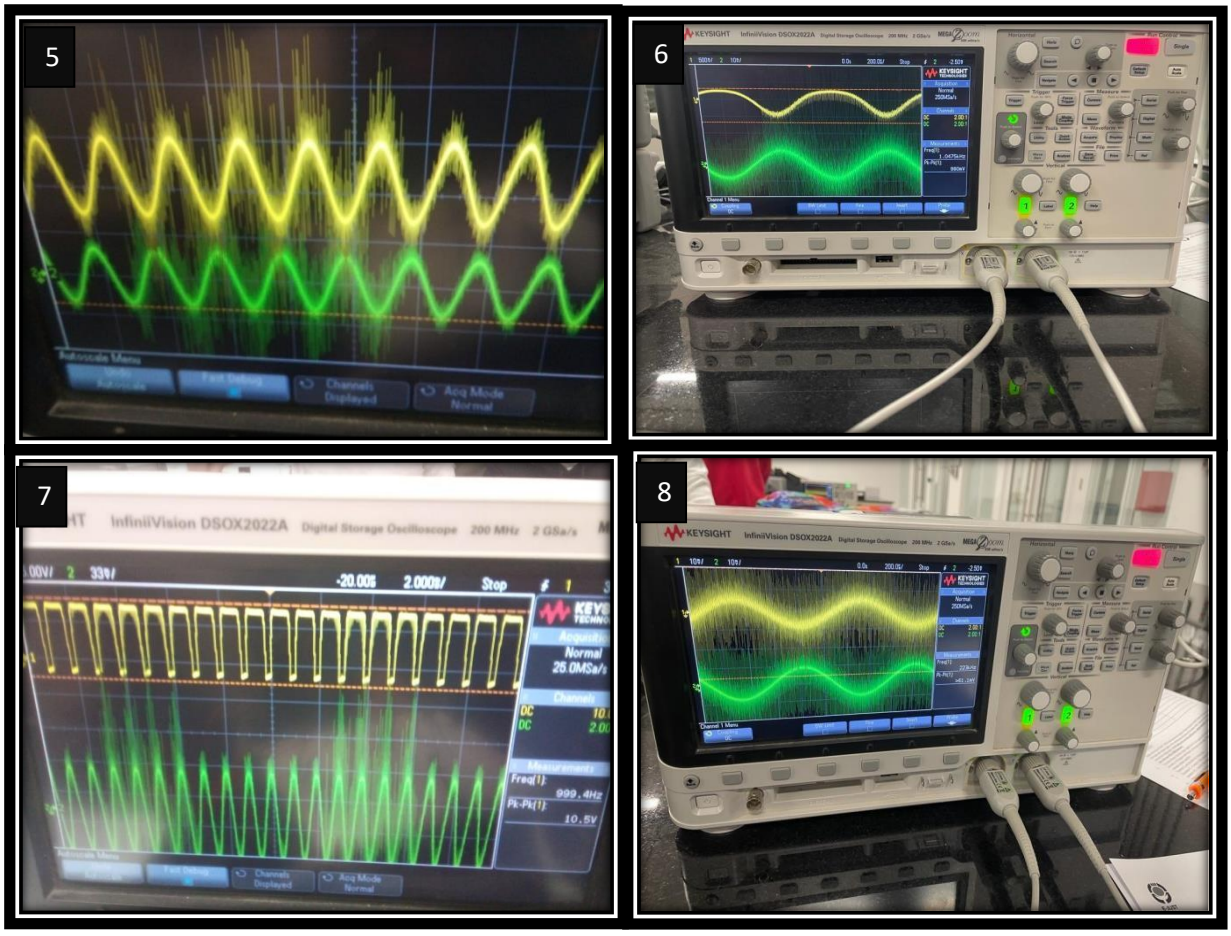
Table 1 Measurements of DC analysis

Parameters	Measured Value	Simulation Value
V_B	3V	3.003V
V_E	2.278V	2.346V
V_C	6.484V	6.49V
I_B	3.6µA	3.43µA
I_E	1.036mA	1.066mA
I_C	1.06mA	1.063mA

$$\beta = \frac{Ic}{IB} = \frac{1.036\text{mA}}{3.6\mu\text{A}} = 377.77$$

$$\alpha = \frac{\beta}{\beta+1} = \frac{377.77}{378.77} = 0.9973$$

Q2) steps 5,6,7,8 graphs are shown in figure below. We observed that there is some noise in our graphs and this noise may be an error.



The error percentage is calculated from the equation:

$$error\ percentage = \frac{Measured\ value - \ simulation\ value}{simulation\ value}$$

The table below shows the values of measured and simulated gain values. I calculated the percentage error from the equation above. The negative sign is for the 180-degree shift. Finally, I calculated the R_{in} and R_{out} for both simulated and measured values.

C_E	Rs	R_L	Vs(p-p)	V_{out}(p-p)	Measured gain	Simulation gain	Error percentage
100µF	Without	3.3K	20mV	9.4V	A_1= -50	A_1= -60	16.66%
Without	Without	3.3K	20mV	800 mV	A2= -0.96	A2= -0.75	28%
100µF	2K	3.3K	20mV	8.5V	A_3= -37.5	A_3= -45	16.66%
100µF	Without	Without	20mV	9.5V	A_4= -198	A_4= -115	72.17%

- ✓ **The causes of the error in the experiment may be because:**

1. Error in value of the component.
2. Unexpected error in the connection.
3. Error in measuring.
4. Error in the used multimeter.

From the rules and using experimental values:

Rin = 6 kΩ

Rout = 9.768 kΩ

From the rules and using simulated values:

R_{in} = 6 kΩ

R_{out} = 3.025 kΩ

We can observe that values of Rout are different in prelab and experiment.